

12-Week Data Scientist Study Tracker

Month 1: Foundations

Week 1 - Python for Data Science

- Learn NumPy: arrays, indexing, math ops
- Learn Pandas: DataFrames, Series, groupby
- Learn visualization: Matplotlib & Seaborn
- Mini Project: Titanic Dataset EDA

Week 2 - Statistics & Probability

- Descriptive stats: mean, median, std dev
- Probability: distributions, Bayes theorem
- Hypothesis testing: p-value, t-test
- Practice on real datasets

Week 3 - Exploratory Data Analysis (EDA)

- Handling missing values, outliers
- Distributions, groupby, aggregations
- Heatmap, scatter, boxplot visualizations
- Mini Project: Housing price EDA

Week 4 - SQL for Data Science

- Learn SELECT, WHERE, JOIN, GROUP BY
- Practice on Hackerrank / Mode
- Mini Project: SQL-based sales dashboard

Month 2: Machine Learning Core

Week 5 - Machine Learning Basics

- Learn supervised ML: regression, classification
- Train/test split, evaluation metrics
- Mini Project: Linear regression model

Week 6 - Tree-based Models

- Decision Trees and Random Forest

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- Confusion matrix, precision, recall, ROC-AUC
- Project: Churn prediction

Week 7 - Unsupervised Learning

- Clustering: KMeans, DBSCAN
- Dimensionality Reduction: PCA
- Project: Customer segmentation

Week 8 - Feature Engineering & Pipelines

- Scaling, encoding, feature selection
- Sklearn pipelines, GridSearchCV
- Project: End-to-end ML pipeline

Month 3: Real-World Projects & Job Prep

Week 9 - Deep Learning Basics (Optional)

- Understand neural nets, activations
- Build basic model in TensorFlow/Keras
- Project: MNIST digit classification

Week 10 - Deployment & Git

- Use GitHub to host code
- Build Flask/Streamlit app
- Project: Deploy ML model to Render/Heroku

Week 11 - Resume & Portfolio

- Build resume, update GitHub & LinkedIn
- Add 3+ projects with READMEs
- Write Medium blog on 1 project

Week 12 - Interview Prep

- Revise Python, Stats, ML Q&A
- Do mock interviews
- Apply to 10+ internships/jobs

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Advanced Data Science Roadmap

Step-by-Step Plan to Become a Professional Data Scientist

Phase 1: Review & Strengthen the Basics (2-3 weeks)

Key Topics:

- Python (data structures, OOP, file I/O)
- NumPy, Pandas, Matplotlib, Seaborn
- SQL (joins, group by, subqueries)
- Scikit-learn: basic models

Practice:

- HackerRank (SQL + Python)
- LeetCode (Easy)
- Kaggle Titanic project

Phase 2: Intermediate ML & Projects (3-4 weeks)

Topics:

- Feature engineering, model evaluation, cross-validation
- Hyperparameter tuning: GridSearchCV, RandomSearch

Projects:

- House price prediction
- Customer churn detection
- Sentiment analysis

Resources:

- "Hands-On ML with Scikit-Learn, Keras & TF"
- Kaggle's Intermediate ML course

Phase 3: Deep Dive into Deep Learning (4-6 weeks)

Advanced Data Science Roadmap

Topics:

- Neural networks, CNNs, RNNs, LSTMs
- TensorFlow or PyTorch

Projects:

- MNIST digit recognition
- Text sentiment classifier
- Image classifier

Resources:

- DeepLearning.AI TensorFlow Specialization
- PyTorch tutorials

Phase 4: Special Topics & Tools (4 weeks)

Topics:

- Time Series Forecasting, NLP
- Streamlit, Flask, Docker, Spark, Cloud (AWS/GCP)

Resources:

- Coursera NLP Specialization
- Kaggle Time Series Course

Phase 5: Career Prep & Real-World Exposure (4+ weeks)

Tasks:

- GitHub portfolio with 3-5 projects
- Technical blogging (Medium/Hashnode)
- Kaggle competitions, apply for internships

Learn:

- Behavioral & technical interview prep

Advanced Data Science Roadmap

- SQL + ML case studies

Weekly Schedule Example

Mon-Wed: Learn new topics (videos, articles)

Thu-Fri: Practice (coding, notebooks)

Sat: Project work

Sun: Revise, GitHub/blog updates